kajall120 / **Data_science_Lab_SE_64** Public

<> **Code** · Issues · Pull requests · Actions · Projects · Security · Insights

Files

main

Go to file

- A_.ipynb
- BFS&DFS.ipynb
- Decisionmaking.ipynb
- Even_odd.ipynb
- MINMAX.ipynb
- README.md
- Untitled2.ipynb
- Untitled5.ipynb
- Untitled6.ipynb
- a.ipynb
- evenodd.ipynb
- expertsystem.ipynb
- factorial.ipynb
- linearequation.ipynb
- meanmedianwith_function.ipynb
- palindromnumber.ipynb

Data_science_Lab_SE_64 / **linearequation.ipynb**

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Preview Code Blame 74 lines (74 loc) · 1.8 KB

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In [ ]:

In [1]:

```python
import numpy as np

# Define the coefficient matrix A
A = np.array([[2, 3],
              [4, 5]])

# Define the constant vector B
B = np.array([10, 20])

# Solve the system of equations
x, y = np.linalg.solve(A, B)

print(f"The solution is: x = {x:.2f}, y = {y:.2f}")
```

```
The solution is: x = 5.00, y = 0.00
```

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- MINMAX.ipynb
- README.md
- Untitled2.ipynb
- Untitled5.ipynb
- Untitled6.ipynb
- a.ipynb
- evenodd.ipynb
- expertsystem.ipynb
- factorial.ipynb
- meanmedianwith_function.ipynb
- palindromnumber.ipynb
- simpleprogram.ipynb
- simpleprogram2.ipynb

Data_Science_Lab_SE_64 / Untitled6.ipynb

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Preview Code Blame 88 lines (88 loc) · 2.24 KB

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In [ ]:

In [2]:

```
import numpy as np

# Define a 3x3 square matrix
matrix = np.array([
    [1, 2, 3],
    [0, 1, 4],
    [5, 6, 0]
])

print("Original 3x3 matrix:")
print(matrix)

# Calculate the inverse of the matrix
try:
    inverse_matrix = np.linalg.inv(matrix)
    print("\nInverse of the matrix:")
    print(inverse_matrix)
except np.linalg.LinAlgError:
    print("\nError: The matrix is singular and does not have an inverse.")
```

```
Original 3x3 matrix:
[[1 2 3]
 [0 1 4]
 [5 6 0]]

Inverse of the matrix:
[[-24.  18.   5.]
 [ 20. -15.  -4.]
 [ -5.   4.   1.]]
```

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Go to file

- A_.ipynb
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- MINMAX.ipynb
- README.md
- Untitled2.ipynb
- Untitled5.ipynb
- a.ipynb
- evenodd.ipynb
- expertsystem.ipynb
- factorial.ipynb
- meanmedianwith_function.ipynb
- palindromnumber.ipynb
- simpleprogram.ipynb
- simpleprogram2.ipynb

Data_Science_Lab_SE_64 / Untitled5.ipynb

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Preview | Code | Blame   98 lines (98 loc) · 2.35 KB

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In [6]:

In [7]:

```
# Define a 3x3 matrix
matrix = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [4, 8, 9]])

print("The 3x3 matrix is:")
print(matrix)
```

```
The 3x3 matrix is:
[[1 2 3]
 [4 5 6]
 [4 8 9]]
```

In [8]:

```
# Compute the determinant of the matrix
determinant = np.linalg.det(matrix)

print(f"\nThe determinant of the matrix is: {determinant}")
```

```
The determinant of the matrix is: 9.000000000000002
```